

535510 Reinforcement Learning

Pre-Lecture Assignment 5

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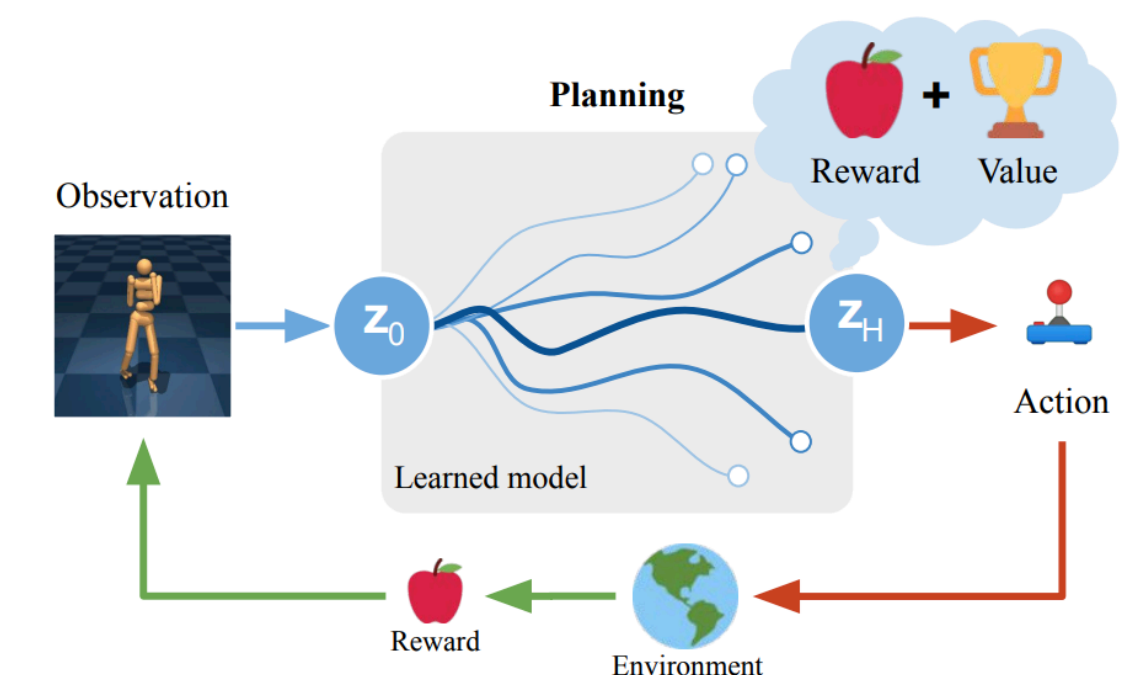
Problem: Model-Based RL

In Lecture 21, we proceed to study Model-based RL methods. Let's try to get familiar with these methods and get prepared for Lecture 22. You can refer to the (1) attached reference slides on model-based RL and the (2) TD-MPC paper (<https://arxiv.org/abs/2203.04955>) as a starting point when answering the questions below.

1) **(Parametric vs non-parametric models)** There are both parametric and non-parametric models. Classify each of the following as **parametric** or **non-parametric**, and briefly justify each of your answer: **(a)** A neural network dynamics model **(b)** A replay buffer used as an empirical transition model **(c)** A symbolic PDDL-style planning model **(d)** A Gaussian process dynamics model **(e)** A table of all observed transitions in a small gridworld

2) **(Model Predictive Control, MPC)** MPC repeatedly plans over a short horizon and executes only the first action.

- (a) Describe the MPC procedure step by step.
- (b) Why does MPC usually execute only the first planned action?
- (c) Give one real-world example of a task where MPC would be useful.
- (d) What's the main difference between vanilla MPC and TD-MPC? (Please mention at least 5 enhancements made by TD-MPC)



Submission Guidelines and Remarks

- Please write all your responses within **1 page (A4 paper)**, either hand-written or typeset). The responses that go beyond 1 page will *not* be graded.
- Your deliverable shall be a PDF file. Please put all your write-ups into one PDF file (photos/scanned copies are acceptable) and submit your deliverable via E3.
- Please submit it by **9pm, 5/28 (Thursday)**.

